① On-Chain Query
User Contract
Oracle Contract
q = (o, r, θ)
object,type,params
e(q)
Emit event
② Dispatch to Oracle Network
Oracle Nodes
③ Hybrid Evidence Collection by Multi-Agents
Oracle Network (Semantic Layer)
Agent-1 (LLM)
Agent-2 (LLM)
Agent-3 (LLM)
Agent-4 (LLM)
Hybrid Evidence Sources
Structured Feeds(Prices, Stats, APIs)
News Articles
Web Pages
Documents(PDF, Reports)
④ Robust Trust Discovery
Source Trust Memory (per source s)
$r_s = \frac{\alpha_s}{\alpha_s + \beta_s}$
Numerical Channel (Value Aggregation)
Numerical reports
$\{x_s^o\}$
Reliability gating
Filter low-trust sources
Weighted median init
Low weight
High weight
Robust start
IRLS with Tukey reweight
estimate
inliers(high weight)
outliers (down-weighted)
Numerical consistency
$c_o^{num} \in [0,1]$
0
0.5
1
deviation from $\hat{x}_0$
Textual Channel (Claim Selection)
Textual claims
$\{t_i^s\}$
SBERT embeddings
Numeric extraction
12
47
3.5
...
Hybrid affinity matrix
$A_o^{(i,j)} = \alpha \cdot \text{sim} + (1-\alpha) \cdot num$
Trust-aware soft voting
$\omega_1\ \omega_2 \ldots \omega_m$
weights Proportional to r_s
Textual consistency
$c_o^{txt} \in [0,1]$
0
0.5
1
claims
Trusted output
Cross-Batch Reliability Update
Numerical consistency
c_s^{num}
1
0.5
0
1
2
3
4
5
Textual consistency
c_s^{txt}
src.
0.82
0.56
0.43
0.41
Object-level consistency
c_s^{obj}
B
Epoch-level consistency
$\bar{c_s}$
0
0.5
1
Update source reliability
(α_s, β_s)
$\alpha_s \leftarrow \alpha_s + \eta \bar{c_s}$
$\beta_s \leftarrow \beta_s + \eta(1 - \bar{c_s})$
$r_s = \frac{\alpha_s}{\alpha_s + \beta_s}$
⑤ Zero-Knowledge Proof
Circuit Decomposition
Overall Circom Circuit
$C = C_{NUM} \land C_{TXT} \land C_{REL}$
C_{NUM}
Numerical reports
Reliability gating
Robust numerical estimation
Numerical consistency
C_{TXT}
SBERT embeddings
Hybrid affinity matrix
Trust-aware voting
Textual consistency
C_{REL}
Numerical consistency
Textual consistency
Object-level consistency
Epoch-level consistency
Public Input and Witness
Public input
$pub = (\{\hat{x}_o, \hat{t}_o\}_{o \in B_e}, \{r_s\}_{s \in S})$
Witness
$wit = (agent\ \ reports\ , intermediate\ \ states\ , consistency\ \ scores\ , Beta - Bernoulli\ )$
Trusted outputs $\hat{x}_o, \hat{t}_o$
numeric
textual
Reliabilities r_s
per-source trust scores
Agent reports h_o^s
Intermediate computation trace
Consistency Scores & others
α_s
β_s
$\exists\, wit \quad s.t.\ C(pub, wit) = 1$
Proof Generation
Setup/Backend (Circom+Groth16)
.circom
Trusted setup(CRS)
Construct Witness
agent reports h_0^s
intermediate states
consistency scores
$\alpha_0, \beta_s, \ldots$
wit
Generate Proof (Prover)
pk
proving key
wit
witness
pub
public input
zkSNARK Prover
π
Proof Verification
$b \leftarrow V_{ERIFY} P_{ROOF}(\text{vk}, pub, \pi)$
Valid
Store final trusted result
$R = (\hat{x}_0, \hat{t}_0)$
Update source reliability r_s
Invalid
Reject proof/discard submission
Request resubmission after correction